



Web Programming
Lecture 4

Bootstrap + CSS Positioning

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Responsive Design.

- Grid System (Bootstrap, tailwind)
- CSS Media Queries

Media Queries

- **What are Media Queries?**
 - CSS rules that apply styles based on device characteristics such as screen width, height, or orientation.
- **Common Use Cases:**
 - Adjusting layouts for different screen sizes.
 - Changing font sizes on smaller screens.
 - Hiding or showing elements.



```
@media (max-width: 600px) {
```

```
  body {
```

```
    background-color: lightblue;
```

```
  }
```

```
}
```

```
@media (min-width: 601px) and (max-width: 900px) {
```

```
  body {
```

```
    background-color: purple;
```

```
  }
```

```
}
```

```
@media (orientation: portrait) {
```

```
  img {
```

```
    width: 100%;
```

```
  }
```

```
}
```



```
/* Extra small devices (phones, 600px and down) */  
@media only screen and (max-width: 600px) {...}  
  
/* Small devices (portrait tablets and large phones,  
600px and up) */  
@media only screen and (min-width: 600px) {...}  
  
/* Medium devices (landscape tablets, 768px and up) */  
@media only screen and (min-width: 768px) {...}  
  
/* Large devices (laptops/desktops, 992px and up) */  
@media only screen and (min-width: 992px) {...}  
  
/* Extra large devices (large laptops and desktops,  
1200px and up) */  
@media only screen and (min-width: 1200px) {...}
```



CSS Layout – Position

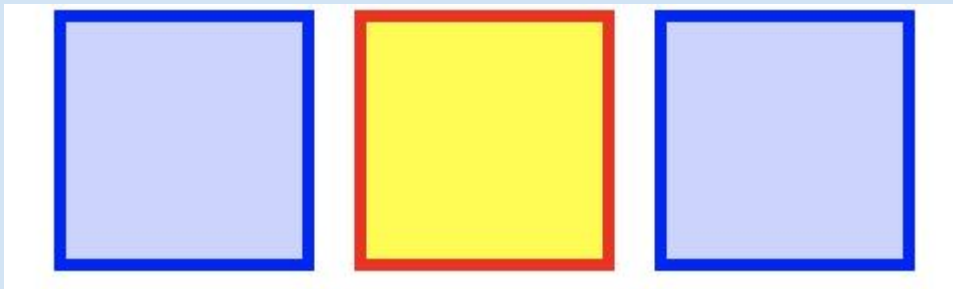
The position is CSS sets how an element is positioned in a document.

- ▶ **Static:** Default positioning; elements are placed in the normal flow. (default)
- ▶ **Relative:** Positioned relative to its normal position.
- ▶ **Absolute:** Positioned relative to the nearest positioned ancestor.
- ▶ **Fixed:** Positioned relative to the viewport; does not move on scroll.
- ▶ **Sticky:** Switches between relative and fixed based on scroll position.

Static



The element is positioned according to the Normal Flow of the document. The top, right, bottom, left, and z-index properties have *no effect*. This is the default value.

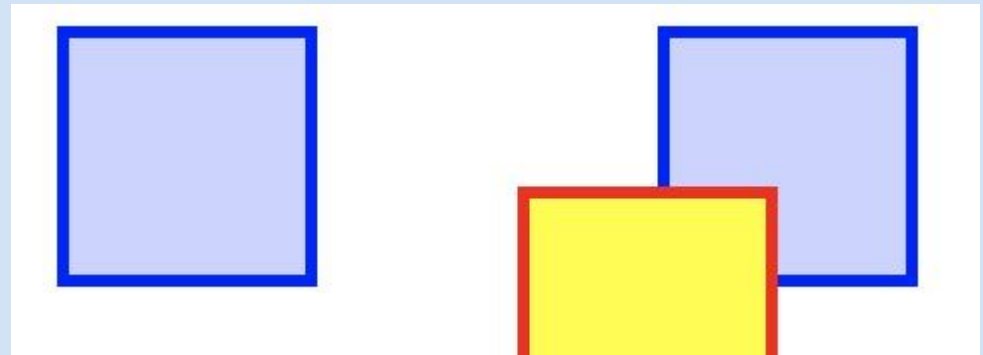


Relative



The element is positioned according to the normal flow of the document, and then offset *relative to itself* based on the values of `top`, `right`, `bottom`, and `left`. The offset does not affect the position of any other elements; thus, the space given for the element in the page layout is the same as if position were static.

`position: relative;`
`top: 40px; left: 40px;`





Absolute

An element with `position: absolute;` is positioned relative to the nearest positioned ancestor, Any ancestor with **position NOT equal to static**: If not ancestor is relative? The absolutely positioned element is positioned **relative to the initial containing block** (usually the **page**)

```
div.relative {  
  position: relative;  
  width: 400px;  
  height: 200px;  
  border: 3px solid green;  
}
```

```
div.absolute {  
  position: absolute;  
  top: 80px;  
  right: 0;  
  width: 200px;  
  height: 100px;  
  border: 3px solid red;  
}
```

This div element has position: relative;

This div element has position: absolute;



Fixed & Sticky

- ▶ `position: fixed`: The element is removed from the normal document flow. Its final position is determined by the values of top, right, bottom, and left. For instance a website with a header and navigation that always stays visible when you are scrolling is a good example of fixed positioning.
- ▶ `position: sticky` basically acts like `position: relative` until an element is scrolled beyond a specific offset, in which case it turns into `position: fixed`

https://www.w3schools.com/css/tryit.asp?filename=trycss_position_sticky

Using position: fixed;

An element with position: fixed; is positioned relative to the viewport, which means it always stays in the same place even if the page is scrolled:

```
div.fixed {  
  position: fixed;  
  bottom: 0;  
  right: 0;  
  width: 300px;  
  height: 200px;  
  border: 3px solid  
}
```

This div element has position: fixed;



What is bootstrap?

- Bootstrap is a powerful, open-source front-end framework used for designing responsive and mobile-first websites.
- Why use bootstrap?
 - Speeds up development.
 - Pre-styled components.
 - Easy to create responsive designs.

Key Features of Bootstrap



- **Responsive Grid System**
- **Pre-designed Components**
 - Buttons, Modals, Alerts, etc.
- **Customizable**
 - Built with Sass and a theming system.
- **Cross-Browser Compatibility**
- **Extensive Documentation**

Bootstrap 5



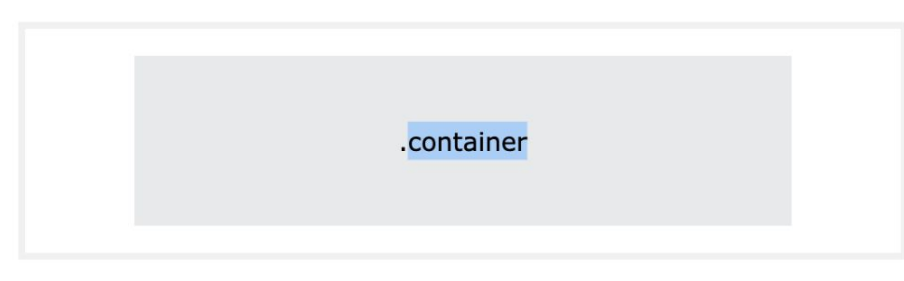
```
<!-- Bootstrap CSS --> <link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/  
bootstrap.min.css" rel="stylesheet"  
integrity="sha384-EVSTQN3/azprG1Anm3QDgpJLIm9Nao0Yz1ztcQTWFs  
pd3yD65VohhpuuCOMLASjC" crossorigin="anonymous">
```

```
<!-- Bootstrap JS Bundle with Popper --> <script  
src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bo  
otstrap.bundle.min.js"  
integrity="sha384-MrcW6ZMFYIzcla8NI+NtUVF0sA7MsXsP1UyJoMp4YL  
EuNSfAP+JcXn/tWtlaxVXM" crossorigin="anonymous"></script>
```

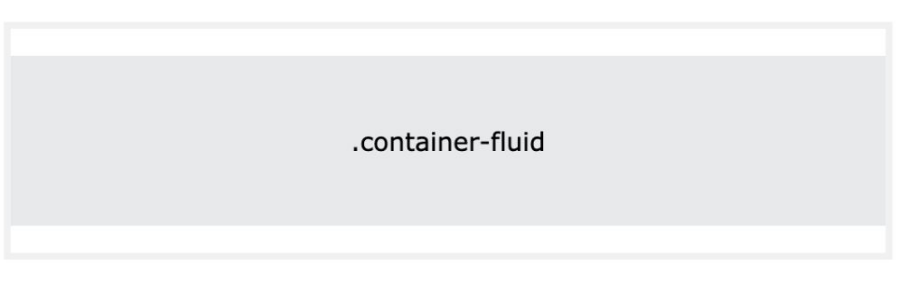
Bootstrap Container

Containers are used to pad the content inside of them, and there are two container classes available:

1. The `.container` class provides a responsive fixed width container
2. The `.container-fluid` class provides a full width container, spanning the entire width of the viewport



`.container`



`.container-fluid`



- | | | | | | | | | | | | |
|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 |
| span 4 | | | | span 4 | | | | span 4 | | | |
| span 4 | | | | span 8 | | | | | | | |
| span 6 | | | | | | span 6 | | | | | |
| span 12 | | | | | | | | | | | |



```
<div class="container">  
  <div class="row">  
    <div class="col-md-6"></div>  
    <div class="col-md-6"></div>  
  </div>  
  <div class="row">  
    <div class="col-md-4"></div>  
    <div class="col-md-4"></div>  
    <div class="col-md-4"></div>  
  </div>  
</div>
```

Typography and Components



- **Typography**
 - Predefined styles for headings, paragraphs, and text utilities.
- **Common Components:**
 - Buttons
 - Alerts
 - Navigation Bars
 - Modals
- **Example:**

```
<button class="btn btn-primary">Click Me!</button>
```



Bootstrap Buttons

Primary

Secondary

Success

Danger

Warning

Info

Light

Dark

Link

```
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-dark">Dark</button>

<button type="button" class="btn btn-link">Link</button>
```

Bootstrap Alerts



```
<div class="alert alert-primary" role="alert">
  This is a primary alert—check it out!
</div>
<div class="alert alert-secondary" role="alert">
  This is a secondary alert—check it out!
</div>
<div class="alert alert-success" role="alert">
  This is a success alert—check it out!
</div>
<div class="alert alert-danger" role="alert">
  This is a danger alert—check it out!
</div>
<div class="alert alert-warning" role="alert">
  This is a warning alert—check it out!
</div>
<div class="alert alert-info" role="alert">
  This is a info alert—check it out!
</div>
<div class="alert alert-light" role="alert">
  This is a light alert—check it out!
</div>
<div class="alert alert-dark" role="alert">
  This is a dark alert—check it out!
</div>
```

This is a primary alert—check it out!

This is a secondary alert—check it out!

This is a success alert—check it out!

This is a danger alert—check it out!

This is a warning alert—check it out!

This is a info alert—check it out!

This is a light alert—check it out!

This is a dark alert—check it out!



Bootstrap Navigation Bar

Active Link Link Disabled

```
<nav class="nav">
  <a class="nav-link active" href="#">Active</a>
  <a class="nav-link" href="#">Link</a>
  <a class="nav-link" href="#">Link</a>
  <a class="nav-link disabled" href="#">Disabled</a>
</nav>
```

- **.nav** uses **display: flex**
- **flex-column** class on **nav** makes it vertical

Active

Link

Link

Disabled

```
<ul class="nav flex-column">
  <li class="nav-item">
    <a class="nav-link active" href="#">Active</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" href="#">Disabled</a>
  </li>
</ul>
```

Bootstrap Tabs



Active

Link

Link

Disabled

```
<ul class="nav nav-tabs">
  <li class="nav-item">
    <a class="nav-link active" href="#">Active</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" href="#">Disabled</a>
  </li>
</ul>
```




Bootstrap Dropdown Menu

Copy

```
<div class="dropdown show">
  <a class="btn btn-secondary dropdown-toggle" href="#" role="button" id="dropdownMenuLink" data-toggle="dropdown" aria-
    Dropdown link
  </a>

  <div class="dropdown-menu" aria-labelledby="dropdownMenuLink">
    <a class="dropdown-item" href="#">Action</a>
    <a class="dropdown-item" href="#">Another action</a>
    <a class="dropdown-item" href="#">Something else here</a>
  </div>
</div>
```

Dropdown link ▼

Action

Another action

Something else here

Bootstrap Form



Email address

We'll never share your email with anyone else.

Password

☐ Check me out

```
<form>
  <div class="form-group">
    <label for="exampleInputEmail1">Email address</label>
    <input type="email" class="form-control" id="exampleInputEmail1" aria-describedby="emailHelp" placeholder="Enter ema
    <small id="emailHelp" class="form-text text-muted">We'll never share your email with anyone else.</small>
  </div>
  <div class="form-group">
    <label for="exampleInputPassword1">Password</label>
    <input type="password" class="form-control" id="exampleInputPassword1" placeholder="Password">
  </div>
  <div class="form-check">
    <input type="checkbox" class="form-check-input" id="exampleCheck1">
    <label class="form-check-label" for="exampleCheck1">Check me out</label>
  </div>
  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

Copy

Bootstrap Responsive

- **Breakpoints:**
 - `sm, md, lg, xl, xxl`
 - Adapt layout based on screen size.
- **Mobile-First Approach:**
 - Prioritize small screens and scale up.
- **Example:**

```
<div class="col-sm-12 col-md-6 col-lg-4">Column 1</div>
```

```
<div class="col-sm-12 col-md-6 col-lg-4">Column 2</div>
```

```
<div class="col-sm-12 col-md-6 col-lg-4">Column 3</div>
```

```
<div class="col-sm-12 col-md-6 col-lg-4">Column 4</div>
```

Resources to Learn



Slide 15: Resources to Learn More

- Websites:
 - <https://getbootstrap.com/docs/4.0/>
- Platforms:
 - FreeCodeCamp, Codecademy, or W3Schools.



Test



Question:

Write complete HTML code using Bootstrap to create a responsive grid layout with 12 boxes that behaves as follows:

Requirements

On mobile phones (small screens) → show 1 column per row

On small screens (phones in landscape / medium devices) → show 2 columns per row

On laptops → show 6 columns per row

Solution



```
<div class="container">
  <div class="row">
    <div class="col-sm-12 col-md-6 col-lg-2">1 </div>
    <div class="col-sm-12 col-md-6 col-lg-2">2</div>
    <div class="col-sm-12 col-md-6 col-lg-2">3</div>
    <div class="col-sm-12 col-md-6 col-lg-2">4</div>
    ...
    <div class="col-sm-12 col-md-6 col-lg-2">12</div>
  </div>
</div>
```